AERODROP

AN AUTONOMOUS DRONE FOR SEARCH & RESCUE

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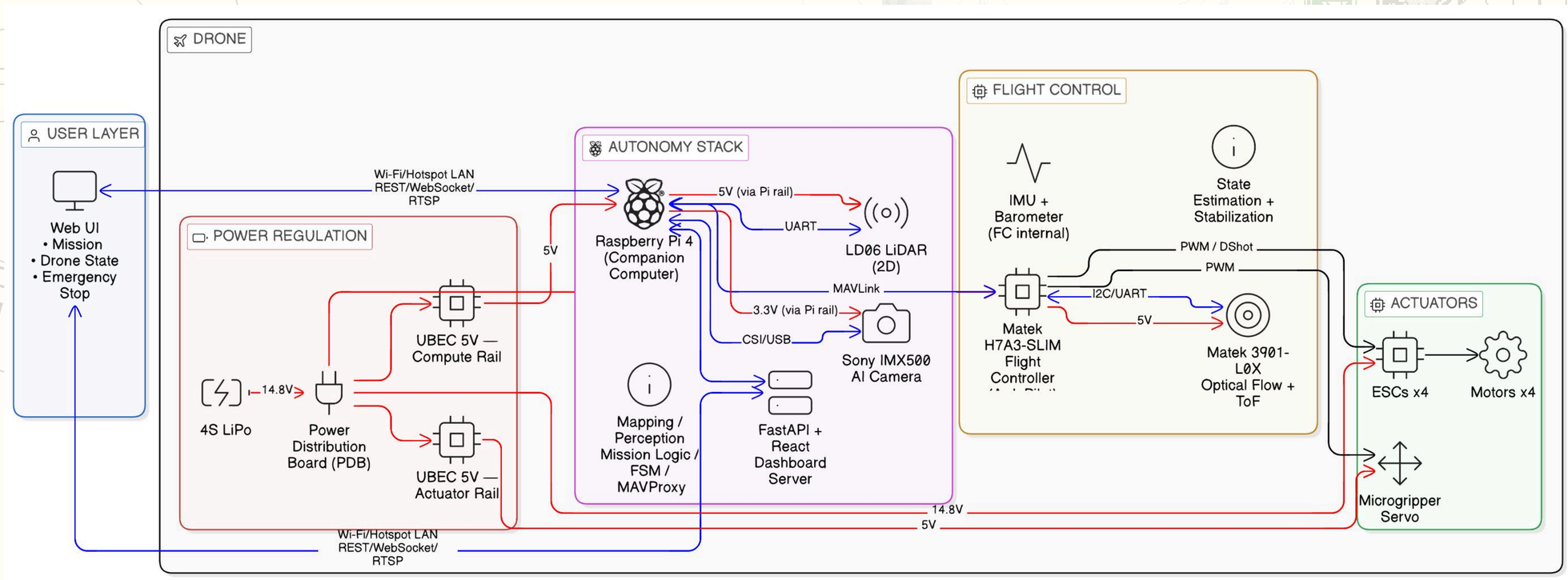
4TH YEAR PROJECT

PROJECT OVERVIEW

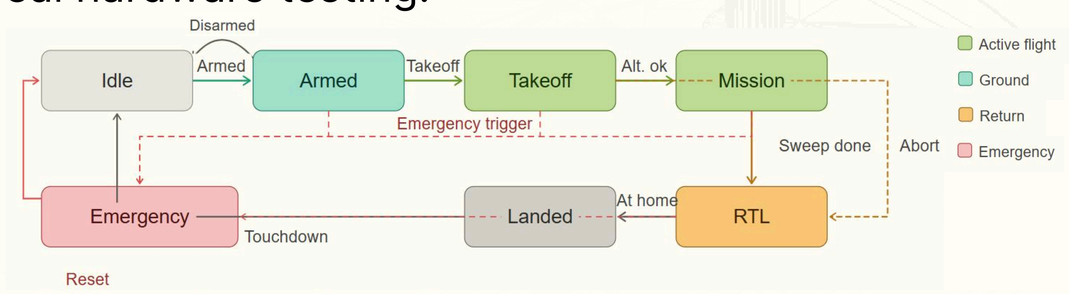
AeroDrop is a fully autonomous quadcopter built from scratch to navigate, map, and deliver payloads inside buildings where GPS does not work. Designed for search and rescue scenarios where ground robots cannot reach and conventional drones lose navigation, it operates without a pilot, without pre-mapped environments, and without any external positioning infrastructure. At its core is a custom-assembled drone running ArduCopter on a Matek flight controller, paired with a Raspberry Pi 4 that handles all mission logic, a 360° LiDAR for live spatial mapping, and an AI camera for target detection at the delivery zone.

ARCHITECTURE DIAGRAM



SOFTWARE

- 7-state Python FSM orchestrates full mission arming to landing driven by live MAVLink telemetry.
- LiDAR scans converted in real time to occupancy grid feeding A* path planner.
- Optical flow + ToF feed EKF3 for stable GPS-denied indoor positioning.
- FastAPI + React dashboard streams telemetry, FSM state, occupancy map, and live video over local hotspot.
- IMX500 AI camera performs on-sensor delivery-zone detection.
- Validated in ArduPilot SITL and HITL simulation before real hardware testing.



HARDWARE

- 4S LiPo → PDB → dual isolated UBEC 5V rails.
- 3901-L0X optical flow + ToF → FC via I2C, fused via EKF3 for GPS-denied position hold.
- LD06 LiDAR → Pi via UART (230400 baud) → occupancy grid → A* planner.
- Pi ↔ FC via UART (115200 baud), pymavlink, commands down, telemetry up.
- IMX500 on-sensor inference → results to Pi via CSI, no frame streaming.
- FC → 4x ESCs via PWM + microgripper servo on actuator rail.

TOOLS & TECH

Websocket	FastAPI	Raspberry Pi OS
ArduCopter 4.6.3	React, Vite	A* Pathfinding
Edge AI Inference	EKF3	ArduPilot SITL, HITL
MAVLink/ pymavlink	MAVSDK	Point Cloud Processing

RESULTS

Built and soldered from scratch, with the full autonomy stack written independently. Validated in ArduPilot SITL and HITL simulation prior to real hardware testing. Stable GPS-denied position hold confirmed, all sensors live, and the end-to-end SAR pipeline runs from arming to payload delivery with real-time dashboard monitoring.

Scan the QR code to view the full results with images and documentation.

RESULTS